

APPLICATION OVERVIEW

Application Name: Discord Quest Manager Version: 1.1.0 Author: Aliester Eroan UI Framework: Tkinter (Python)
Theme System: Catpuccin (Dark/Light) Font System: Press Start 2P (Pixel Font) Layout Style: Retro Terminal / 8-bit Aesthetic

MAIN WINDOW STRUCTURE

Dimensions: 1000 x 650 pixels Resizable: False Background Color: base (#11111b dark / #eff1f5 light) Window
Layout Structure:
- Left Sidebar: 250px width, background=mantle
- Right Main Panel: 750px width, background=base

SIDEBAR COMPONENTS

1. HEADER CARD

Location: Top of sidebar Container: Frame
- Padding: padx=12, pady=15
- Background: surface color
- Outer padding: 10px (all sides)
Components:
- App Icon
* Size: Default application icon size * Location: Left side of header row * Right padding: 10px * Source: IconHandler.load_icon()
- Title Label
* Text: "Discord Quest\nManager" (two lines) * Font: Press Start 2P, size 7 * Color: pink (#f5c2e7 dark / #ea76cb light) * Left padding: 6px * Alignment: Left
- Status Label
* Text: "● DATABASE READY" (dynamic) * Font: Press Start 2P, size 6, italic * Color: green (#a6e3a1 dark / #40a02b light) * Alignment: anchor="w" (left)
- Game Count Label
* Text: "Connecting..." / "X,XXX registered" (dynamic) * Font: Segoe UI, size 8 * Color: subtext0 (#a6adc8 dark / #6c7086 light) * Bottom padding: 15px * Alignment: anchor="w" (left)

2. QUEUE CARD

Location: Below header card Container: Frame

- Padding: padx=15, pady=15
- Background: surface color
- Outer padding: 10px (all sides)

Components:

- Queue Label
 - * Text: "SELECTED: 0 Games Queued" (dynamic) * Font: Press Start 2P, size 6 (badge style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Padding: left 6px, right 5px * Alignment: anchor="w" (left)

3. FAVORITES CARD

Location: Below queue card Container: Frame

- Padding: padx=15, pady=15
- Background: surface color
- Outer padding: 10px (all sides)

Components:

- Favorites Header
 - * Icon: bookmark.png (16x16 pixels) * Text: " Favorites" * Font: Press Start 2P, size 7, bold * Color: text (#cdd6f4 dark / #1e1e2e light) * Icon padding: right 5px * Layout: Horizontal (icon left, text right)
- Favorites Listbox
 - * Height: 5 items (visible) * Font: Press Start 2P, size 6 * Background: surface0 (#313244 dark / #ccd0da light) * Text color: text (#cdd6f4 dark / #1e1e2e light) * Selection background: mauve (#cba6f7 dark / #8839ef light) * Selection foreground: base (#11111b dark / #eff1f5 light) * Border: flat, bd=0, highlightthickness=0 * Event: Double-click to add to queue

4. RECENT CARD

Location: Below favorites card Container: Frame

- Padding: padx=15, pady=15
- Background: surface color
- Outer padding: 10px (all sides)

Components:

- Recent Header
 - * Text: "■ Recent" * Font: Press Start 2P, size 7, bold * Color: text (#cdd6f4 dark / #1e1e2e light) * Bottom padding: 10px * Alignment: anchor="w" (left)
- Recent Listbox
 - * Height: 5 items (visible) * Font: Press Start 2P, size 6 * Background: surface0 (#313244 dark / #ccd0da light) * Text color: text (#cdd6f4 dark / #1e1e2e light) * Selection background: mauve (#cba6f7 dark / #8839ef light) * Selection foreground: base (#11111b dark / #eff1f5 light) * Border: flat, bd=0, highlightthickness=0 * Event: Double-click to add to queue

5. BUTTON CONTAINER (Sidebar Bottom) Location: Bottom of sidebar Container: Frame

- Background: mantle (#1e1e2e dark / #e6e9ef light)

- Padding: padx=10, pady=(5, 20) (top 5, bottom 20, sides 10)
- Layout: side="bottom", fill="x"

Components:

- Clean Dummies Button (PixelButton)
 - * Width: 200 pixels * Height: 45 pixels * Icon: trash.png (18x18 pixels) * Text: "Clean Dummies" * Font: Courier, size 10, bold * Background: surface0 (#313244 dark / #ccd0da light) * Text color: red (#f38ba8 dark / #d20f39 light)
 - * Command: _clean_dummies() * Bottom padding: 5px * Layout: fill="x"
- Settings Button (PixelButton)
 - * Width: 200 pixels * Height: 45 pixels * Icon: cog.png (18x18 pixels) * Text: "Settings" * Font: Courier, size 10, bold
 - * Background: surface0 (#313244 dark / #ccd0da light) * Text color: text (#cdd6f4 dark / #1e1e2e light) *
 - Command: _show_settings_dialog() * Layout: fill="x"

MAIN PANEL COMPONENTS

1. TOOLBAR

- Location: Top of main panel Container: Frame
- Background: mantle (#1e1e2e dark / #e6e9ef light)
 - Padding: padx=15, pady=10
 - Layout: fill="x"

Components:

- Search Frame
 - * Background: surface0 (#313244 dark / #ccd0da light) * Layout: side="left", fill="x", expand=True * Right padding: 10px * Border: flat, bd=0
- Search Icon Label
 - * Icon: search.png (16x16 pixels) * Background: surface0 (#313244 dark / #ccd0da light) * Padding: padx=(10, 5), pady=8 (left 10, right 5, top/bottom 8) * Layout: side="left"
- Search Entry
 - * Font: Press Start 2P, size 7 (entry style) * Background: surface0 (#313244 dark / #ccd0da light) * Text color: text (#cdd6f4 dark / #1e1e2e light) * Cursor color: green (#a6e3a1 dark / #40a02b light) * Padding: padx=8, ipady=5 * Border: flat, bd=0, highlightthickness=0 * Layout: side="left", fill="x", expand=True * Events: KeyRelease, Escape, FocusOut

2. SEARCH DROPDOWN

- Location: Below toolbar (hidden by default) Container: Frame
- Background: surface0 (#313244 dark / #ccd0da light)
 - Layout: place() positioning
 - Visibility: Toggled on search input

3. TARGET FRAME

Location: Center of main panel Container: Frame

- Background: base (#11111b dark / #eff1f5 light)
- Padding: padx=15, pady=10
- Layout: fill="both", expand=True

Components:

- Target Header
 - * Background: base (#11111b dark / #eff1f5 light) * Layout: fill="x", bottom padding 10px
- Title Label
 - * Text: "TARGET EXECUTABLES" * Font: Press Start 2P, size 7, bold (card_title style) * Color: pink (#f5c2e7 dark / #ea76cb light) * Layout: side="left"
- Selected Count Label
 - * Text: "0 Selected" (dynamic) * Font: Press Start 2P, size 6, italic (status style) * Color: subtext0 (#a6adc8 dark / #6c7086 light) * Layout: side="right"

4. EXECUTABLES CONTAINER

Location: Below target header Container: Frame

- Background: mantle (#1e1e2e dark / #e6e9ef light)
- Border: bd=2, highlightbackground=mauve (#cba6f7 dark / #8839ef light), highlightthickness=2
- Layout: fill="both", expand=True

Components:

- Empty State Frame
 - * Background: mantle (#1e1e2e dark / #e6e9ef light) * Layout: fill="both", expand=True * Visibility: Shown when no games in queue
- Folder Icon Label
 - * Icon: folder.png (48x48 pixels) * Background: mantle (#1e1e2e dark / #e6e9ef light) * Position: relx=0.5, y=20, anchor="center" (centered horizontally) * Top padding: 20px
- Empty State Title Label
 - * Text: "NO GAMES ADDED" * Font: Press Start 2P, size 7, bold (empty_state style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Background: mantle (#1e1e2e dark / #e6e9ef light) * Bottom padding: 5px
- Empty State Subtitle Label
 - * Text: "Type in the search bar above to select and queue game executables." * Font: Press Start 2P, size 6 (body_small style) * Color: subtext0 (#a6adc8 dark / #6c7086 light) * Background: mantle (#1e1e2e dark / #e6e9ef light) * Wrap length: 380px
- Results Frame (hidden by default)
 - * Background: mantle (#1e1e2e dark / #e6e9ef light) * Layout: fill="both", expand=True * Visibility: Shown when search results available
- Results Listbox
 - * Font: Press Start 2P, size 6 (listbox style) * Background: surface0 (#313244 dark / #ccd0da light) * Text color: text (#cdd6f4 dark / #1e1e2e light) * Selection background: mauve (#cba6f7 dark / #8839ef light) * Selection foreground: base (#11111b dark / #eff1f5 light) * Border: flat, bd=0, highlightthickness=0 * Selection mode: extended * Scrollbar: Vertical
- 5. QUEUE ITEMS (Dynamic) Location: Inside executables container Container: Frame (per item)
 - Background: surface0 (#313244 dark / #ccd0da light)
 - Padding: padx=5, pady=2

- Layout: fill="x"

Components:

- Checkbox

* Background: surface0 (#313244 dark / #ccd0da light) * Selection color: surface0 (#313244 dark / #ccd0da light) * Left padding: 5px * Command: Toggle queue item checked state

- Game Info Label

* Text: "Game Name → exe_name" (dynamic) * Font: Press Start 2P, size 7 (entry style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Background: surface0 (#313244 dark / #ccd0da light) * Padding: left 5px * Layout: side="left", fill="x", expand=True * Alignment: anchor="w" (left)

- Favorite Button

* Icon: bookmark.png (16x16 pixels) * Background: surface0 (#313244 dark / #ccd0da light) * Right padding: 2px * Layout: side="right" * Command: Toggle favorite status

- Remove Button

* Icon: close.png (16x16 pixels) * Background: surface0 (#313244 dark / #ccd0da light) * Right padding: 2px * Layout: side="right" * Command: Remove from queue

6. FOOTER

Location: Bottom of main panel Container: Frame

- Background: mantle (#1e1e2e dark / #e6e9ef light)

- Padding: padx=15, pady=15

- Layout: fill="x", side="bottom"

Components:

- Timer Label

* Text: "15:00" (dynamic) * Font: Press Start 2P, size 13 (pixel_timer style) * Color: green (#a6e3a1 dark / #40a02b light) * Background: mantle (#1e1e2e dark / #e6e9ef light) * Bottom padding: 10px * Layout: side="left"

- Quest Count Label

* Text: "0 QUESTS READY" (dynamic) * Font: Press Start 2P, size 6, italic (status style) * Color: subtext0 (#a6adc8 dark / #6c7086 light) * Background: mantle (#1e1e2e dark / #e6e9ef light) * Bottom padding: 10px * Layout: side="left"

- Start/Stop Button (PixelButton)

* Width: 220 pixels * Height: 45 pixels * Icon: play.png / stop.png (18x18 pixels, dynamic) * Text: "Start Quests (0)" / "Stop All Quests" (dynamic) * Font: Courier, size 10, bold * Background: green (#a6e3a1 dark / #40a02b light) / red (#f38ba8 dark / #d20f39 light, dynamic) * Text color: base (#11111b dark / #eff1f5 light) * Command: _on_start_stop_button_click() * Bottom padding: 10px * Layout: side="right" * State: disabled when no games selected

7. PROGRESS BAR

Location: Below footer controls Container: Frame

- Background: mantle (#1e1e2e dark / #e6e9ef light)

- Top padding: 10px

- Layout: fill="x"

Components:

- Progress Canvas
 - * Height: 16 pixels * Background: mantle (#1e1e2e dark / #e6e9ef light) * Border: flat, highlightthickness=0 *
 - Layout: fill="x", side="left", expand=True
- Progress Trough (Background)
 - * Height: 16 pixels * Width: 1000 pixels (full width) * Color: mantle (#1e1e2e dark / #e6e9ef light) * Border: none
- Progress Fill (Foreground)
 - * Height: 16 pixels * Width: 0 pixels (dynamic, based on quest progress) * Color: green (#a6e3a1 dark / #40a02b light) * Border: none

SETTINGS DIALOG COMPONENTS

Window Properties:

- Dimensions: 400 x 450 pixels
- Resizable: False
- Background: base (#11111b dark / #eff1f5 light)
- Title: "Discord Quest Manager - Settings"
- 1. NOTEBOOK (Tab Container) Container: ttk.Notebook
- Padding: padx=10, pady=10
- Layout: fill="both", expand=True

Tabs:

- Theme Tab
- Quest Tab
- System Tab
- Updates & About Tab

2. THEME TAB

Container: Frame

- Background: base (#11111b dark / #eff1f5 light)

Components:

- Theme Label
 - * Text: "Theme:" * Font: Press Start 2P, size 7, bold (label style) * Color: text (#cdd6f4 dark / #1e1e2e light) *
 - Background: base (#11111b dark / #eff1f5 light) * Padding: left 10px, top 10px, bottom 5px * Alignment: anchor="w" (left)
- Theme Frame
 - * Background: base (#11111b dark / #eff1f5 light) * Padding: left 10px, top 5px, bottom 5px * Layout: fill="x"
- Dark Theme Radio Button
 - * Icon: full-moon.png (16x16 pixels) * Padding: right 5px * Text: "Catpuccin Mocha (Dark)" * Font: Press Start 2P, size 6 (about_text style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Background: base (#11111b dark / #eff1f5 light) * Selection color: surface0 (#313244 dark / #ccd0da light) * Active background: base (#11111b dark / #eff1f5 light) * Value: "dark" * Layout: anchor="w"
- Light Theme Radio Button

* Icon: sun.png (16x16 pixels) * Padding: right 5px * Text: "Catppuccin Latte (Light)" * Font: Press Start 2P, size 6 (about_text style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Background: base (#11111b dark / #eff1f5 light) * Selection color: surface0 (#313244 dark / #ccd0da light) * Active background: base (#11111b dark / #eff1f5 light) * Value: "light" * Layout: anchor="w"

3. QUEST TAB

Container: Frame

- Background: base (#11111b dark / #eff1f5 light)

Components:

- Duration Label

* Text: "Quest Duration (minutes):" * Font: Press Start 2P, size 7, bold (label style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Background: base (#11111b dark / #eff1f5 light) * Padding: left 10px, top 10px, bottom 5px * Alignment: anchor="w" (left)

- Preset Frame

* Background: base (#11111b dark / #eff1f5 light) * Padding: left 10px, top 5px, bottom 5px * Layout: fill="x"

- Preset Buttons (4 buttons)

* Text: "5", "10", "15", "30" * Font: Press Start 2P, size 6, bold (button_small style) * Background: surface1 (#45475a dark / #bcc0cc light) * Text color: text (#cdd6f4 dark / #1e1e2e light) * Width: 5 characters * Padding: left 2px, right 2px * Layout: side="left" * Command: Set duration to button value

- Duration Entry

* Font: Press Start 2P, size 7 (entry style) * Background: surface0 (#313244 dark / #ccd0da light) * Text color: text (#cdd6f4 dark / #1e1e2e light) * Cursor color: white * Width: 8 characters * Left padding: 5px * Layout: side="left" * Border: flat, bd=0

- Remember Duration Checkbox

* Text: "Remember last used duration" * Font: Press Start 2P, size 6 (about_text style) * Color: text (#cdd6f4 dark / #1e1e2e light) * Background: base (#11111b dark / #eff1f5 light) * Selection color: surface0 (#313244 dark / #ccd0da light) * Active background: base (#11111b dark / #eff1f5 light) * Padding: left 10px, top 10px, bottom 10px * Alignment: anchor="w" (left)

4. SYSTEM TAB

Container: Frame

- Background: base (#11111b dark / #eff1f5 light)

- Components vary based on system settings

5. UPDATES & ABOUT TAB

Container: Frame

- Background: base (#11111b dark / #eff1f5 light)

- Components vary based on update status

6. SAVE BUTTON

Location: Bottom of dialog Container: Button

- Text: "Save Settings"

- Font: Press Start 2P, size 7 (button style)
- Background: green (#a6e3a1 dark / #40a02b light)
- Text color: mantle (#1e1e2e dark / #e6e9ef light)
- Padding: pady=5, padx=20
- Top padding: 10px
- Layout: pack()
- Command: `_save_settings()`

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PIXELBUTTON CUSTOM COMPONENT

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Component Type: Custom Canvas Widget File: `ui/pixel_button.py`

Properties:

- Width: Configurable (default 160px, used: 200px sidebar, 220px footer)
- Height: Configurable (default 50px, used: 45px)
- Background Color: Configurable (surface0, green, red)
- Text Color: Configurable (text, red, base)
- Icon: Optional PhotoImage (18x18 pixels)
- Font: Courier, size 10, bold
- State: normal / disabled

Visual Features:

- Pixelated rounded corners using staircasing technique
- Pixel scale: 3 (for corner stepping)
- 3D shading with light/dark borders
- Hover effect with color lightening
- Press effect with color darkening and offset
- Disabled state with darkened color

Icon/Text Layout:

- Icon position: $\text{pixel_scale} * 4 + 4$ (left offset)
- Icon anchor: center
- Text position: $\text{icon_x} + 16$ (16px right of icon)
- Text anchor: w (left-aligned)
- Text vertical position: $\text{height} // 2 + 2$ (when pressed)

Color Calculations:

- Hover color: `lighten(bg_color, 15)`
- Shadow color: `darken(bg_color, 20)`
- Disabled color: `darken(bg_color, 40)`

Events:

- Mouse Enter: Show hover color
- Mouse Leave: Return to base color
- Mouse Press: Show shadow color with offset

- Mouse Release: Return to hover color and execute command

FONT REFERENCE TABLE

Font Name Size Style Usage

Press Start 2P 7 - App title, section titles, buttons Press Start 2P 6 italic Status labels, game count Press Start 2P 6
- Listbox items, body text Press Start 2P 6 bold Badges, small buttons Press Start 2P 7 bold Labels, card titles
Press Start 2P 13 - Timer display Press Start 2P 5 - Footer text Courier 10 bold PixelButton text Segoe UI 8 -
Game count label

Font Key Constants (FONTS dict):

- app_title: (Press Start 2P, 7)
- section_title: (Press Start 2P, 7)
- badge: (Press Start 2P, 6)
- item_label: (Press Start 2P, 7)
- body_small: (Press Start 2P, 6)
- button: (Press Start 2P, 7)
- footer: (Press Start 2P, 5)
- timer: (Press Start 2P, 13)
- header: (Press Start 2P, 7)
- status: (Press Start 2P, 6, italic)
- label: (Press Start 2P, 7, bold)
- entry: (Press Start 2P, 7)
- listbox: (Press Start 2P, 6)
- button_small: (Press Start 2P, 6, bold)
- about_title: (Press Start 2P, 7, bold)
- about_version: (Press Start 2P, 6, italic)
- about_text: (Press Start 2P, 6)
- pixel_timer: (Press Start 2P, 13, bold)
- pixel_timer_small: (Press Start 2P, 10, bold)
- card_title: (Press Start 2P, 7, bold)
- empty_state: (Press Start 2P, 7, bold)
- dropdown: (Press Start 2P, 7)
- body: (Press Start 2P, 6)

ICON REFERENCE TABLE

Icon Name Size Usage

app_icon Default Window icon, header app icon bookmark.png 16x16 Favorites header, queue favorite button
trash.png 18x18 Clean Dummies button cog.png 18x18 Settings button search.png 16x16 Search bar folder.png

48x48 Empty state placeholder play.png 18x18 Start button (PixelButton) stop.png 18x18 Stop button (PixelButton)
close.png 16x16 Queue remove button sun.png 16x16 Light theme radio button full-moon.png 16x16 Dark theme
radio button

Icon Loading:

- UI icons: IconHandler.load_ui_icon(filename, size_tuple)
- App icon: IconHandler.load_icon()
- All icons maintain reference to prevent garbage collection

COLOR REFERENCE TABLE

Color Name Dark Theme Light Theme Usage

base #11111b #eff1f5 Main window background mantle #1e1e2e #e6e9ef Sidebar, toolbar, footer surface #252538
#e6e9ef Card containers surface0 #313244 #ccd0da Inputs, listboxes, button backgrounds surface1 #45475a
#bcc0cc Borders, preset buttons surface2 #585b70 #acb0be Hover states text #cdd6f4 #1e1e2e Primary text
subtext0 #a6adc8 #6c7086 Secondary text, muted labels subtext1 #bac2de #5c5f77 Tertiary text overlay0
#6c7086 #9ca0b0 Overlay backgrounds overlay1 #7f849c #8c8fa1 Overlay borders overlay2 #9399b2 #7c7f93
Overlay highlights blue #89b4fa #1e66f5 Blue accents lavender #b4befe #7287fd Lavender accents sapphire
#74c7ec #209fb5 Sapphire accents sky #89dceb #04a5e5 Sky accents teal #94e2d5 #179299 Teal accents green
#a6e3a1 #40a02b Status, success, start button yellow #f9e2af #df8e1d Yellow accents peach #fab387 #fe640b
Peach accents maroon #eba0ac #e64553 Maroon accents red #f38ba8 #d20f39 Error, stop button, clean dummies
mauve #cba6f7 #8839ef Selection, highlights, borders pink #f5c2e7 #ea76cb Titles, accents flamingo #f2cddc
#dd7878 Flamingo accents rosewater #f5e0dc #dc8a78 Rosewater accents

Color Constants:

- COLORS_DARK: Dictionary of dark theme colors (Catppuccin Mocha)
- COLORS_LIGHT: Dictionary of light theme colors (Catppuccin Latte)
- COLORS: Default to COLORS_DARK
- Theme switching handled by ThemeManager

LAYOUT SPACING REFERENCE

Standard Padding Values:

- Card outer padding: 10px (all sides)
- Card inner padding: 12-15px (sides), 15px (top/bottom)
- Button container padding: 10px (sides), 5-20px (top/bottom)
- Toolbar padding: 15px (sides), 10px (top/bottom)
- Target frame padding: 15px (sides), 10px (top/bottom)
- Footer padding: 15px (sides), 15px (top/bottom)
- Icon padding: 5-10px (horizontal), 8px (vertical for search)
- Text padding: 5-6px (horizontal), 2-15px (vertical)

Standard Dimensions:

- Sidebar width: 250px
- Main panel width: 750px
- PixelButton width: 200px (sidebar), 220px (footer)
- PixelButton height: 45px
- Icon sizes: 16x16 (small), 18x18 (buttons), 48x48 (large)
- Progress bar height: 16px
- Listbox height: 5 items (visible)

Standard Border Widths:

- Executables container: 2px
- Highlight thickness: 2px
- Button borders: 0px (flat)
- Input borders: 0px (flat)

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UI BEHAVIOR REFERENCE

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Dynamic Text Updates:

- Status label: "● DATABASE READY" / "● DATABASE OFFLINE"
- Game count: "Connecting..." / "X,XXX registered"
- Queue label: "SELECTED: X Games Queued"
- Selected count: "X Selected"
- Quest count: "X QUESTS READY"
- Start button: "Start Quests (X)" / "Stop All Quests"
- Timer: "MM:SS" format

State Management:

- Button states: normal / disabled
- Theme: dark / light (switchable)
- Quest states: idle / running
- Database states: loading / ready / offline

Event Handling:

- Double-click: Add to queue (favorites/recent)
- Single-click: Select (listbox)
- Hover: Color changes (PixelButton)
- Press: Offset and color change (PixelButton)
- Release: Execute command (PixelButton)

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END OF REFERENCE

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